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BAHRAIN POLYTECHNIC

AWS Academy Data Center Technician

Module 7: Power Supplies

aws academy

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Module overview

Lessons

1. Identify use case of a PSU
2. Install and remove a PSU
3. Troubleshoot a PSU



Knowledge check

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Module objectives

At the end of this module, you should be able to:

- Identify the purpose of a power supply unit (PSU)
- Identify the physical elements of a PSU
- Identify the steps to remove a PSU from a PC
- Identify the steps to install a PSU in a PC
- Identify the symptoms of PSU-related issues
- Identify troubleshooting strategies for PSU-related issues

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Module 7: Power Supplies

Lesson 1: Identify use case of a PSU

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Purpose

A power supply unit (PSU) converts high voltage alternating current (AC) drawn from the power mains into low voltage direct current (DC).

DC is required to **operate** computer circuits and certain mechanical elements.

It is important to maintain a constant supply of electricity.

This is needed to generate, transmit, and maintain the electrical signal pulses required to operate the electronic and mechanical elements of a computer.

A PSU is sometimes called a **switched-mode power supply (SMPS)**.



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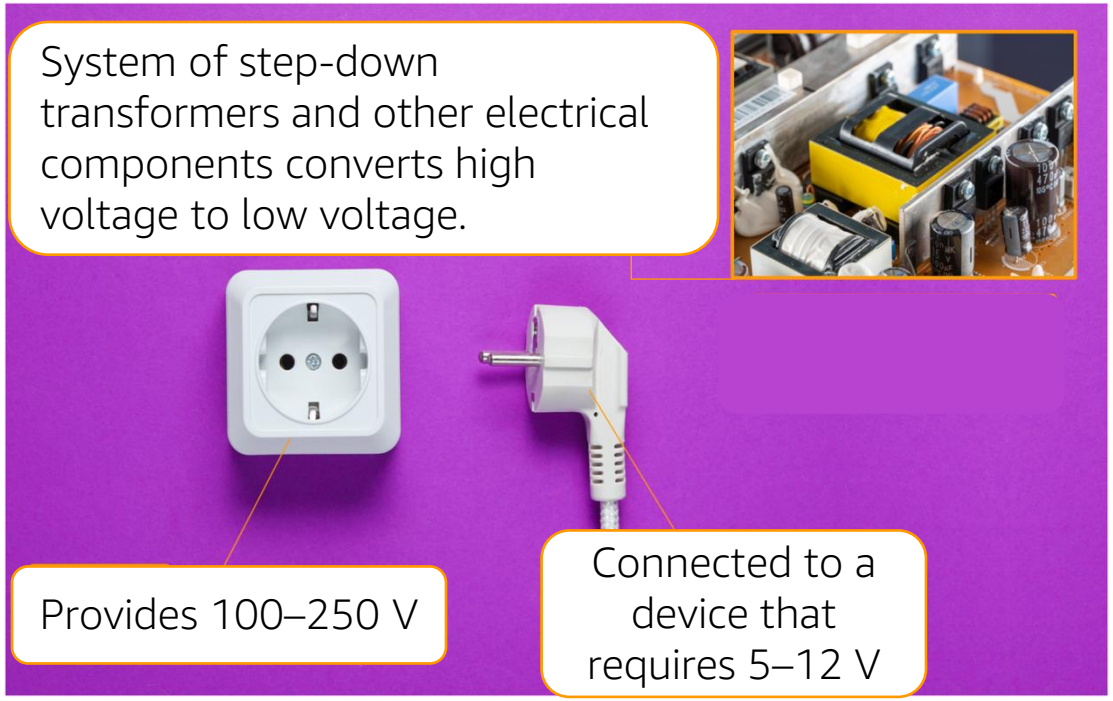
Unit voltages

The voltage of the AC power supply available in homes and offices is typically 100–250 V, depending on geography and electricity standards.

This **high voltage power needs to be converted to the low voltage level** required to operate DC-dependent devices, which is 5-12 V.

To do this, step down transformers and an arrangement of other electrical components are typically used.

Input voltages vary by geography and local electrical standards. The regulator must be set to the right input voltage for the PSU to function correctly.



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Evolution



PSUs were used well before personal computers (PCs) became popular.

Devices such as radio receivers, televisions, and gaming consoles use different PSU types to manage power requirements.

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Evolution (cont.)



The first dedicated, built-in PSU used inside a PC was designed for the Apple II by Atari engineer, Rod Holt.

Before this time, cases, PSUs, RF converters, and RF-converter PSUs were sold separately.

These elements were used to connect the computer board to a home television set.

The keyboard was sold separately too.

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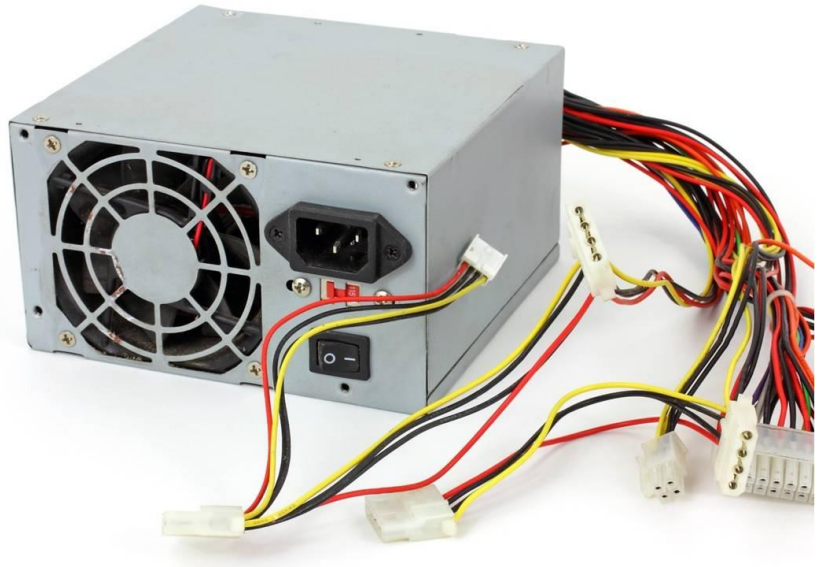
Evolution (cont.)

Early computers operated using combinations of DC +/-3.3 V, +/-5 V, and +/-12 V power.

Different voltages were used to power different elements of the computer or the attached peripheral devices.

In modern computers:

- A +12 V supply is used for cooling fans, hard disk drives (HDDs), serial interfaces, and high-powered graphics processing units (GPUs).
- +3.3 V and +5 V supplies are used for almost all other hardware components, such as central processing units (CPUs), random access memory (RAM), and the motherboard circuitry.



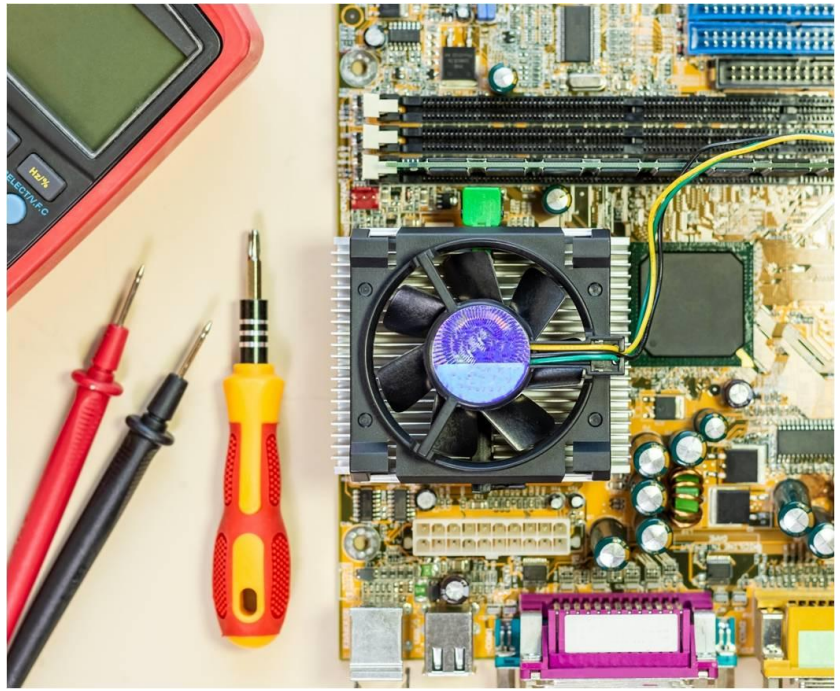
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Power rating

Power rating is the **power consumption drawn from the power main.**

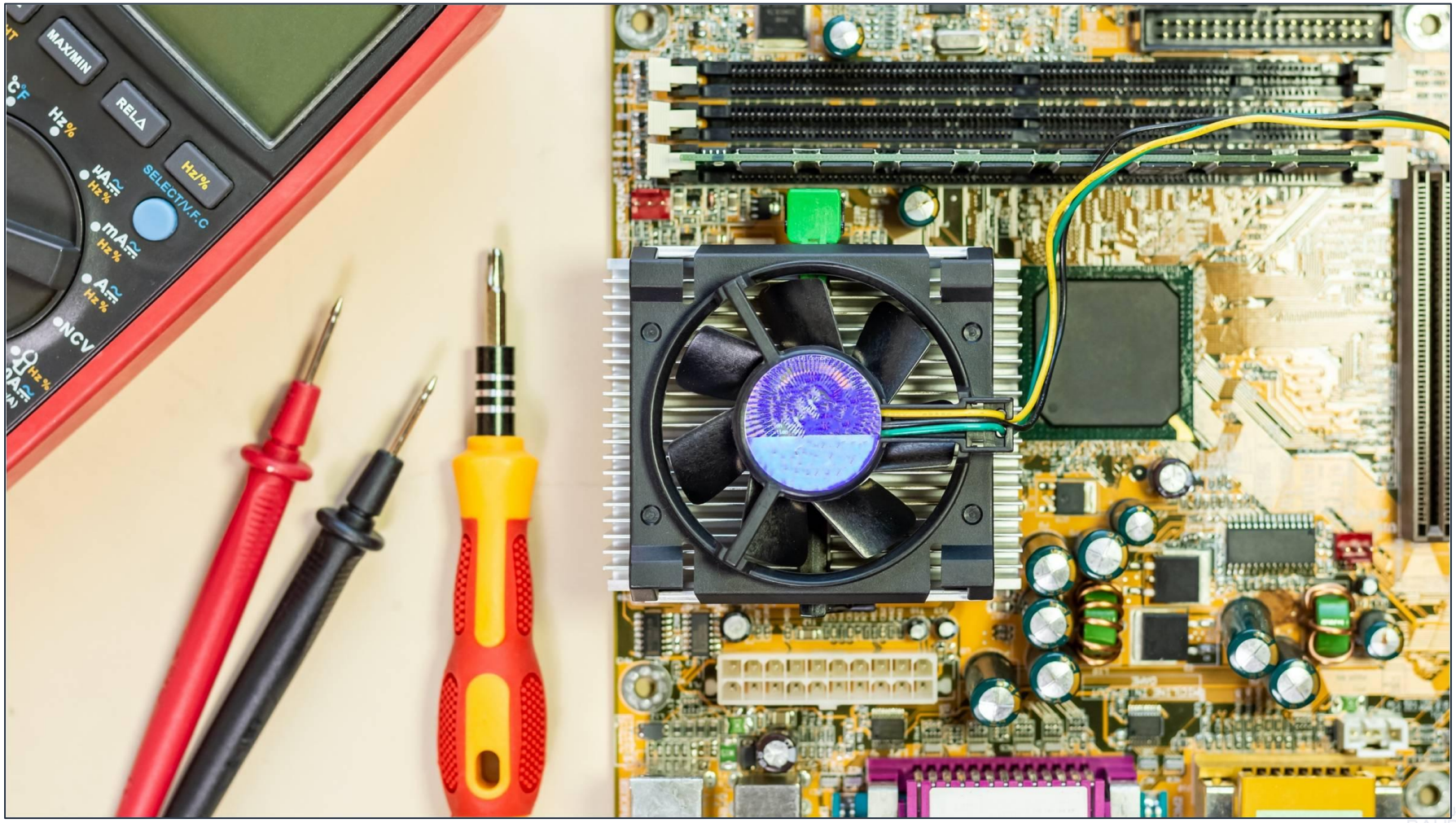
Because all power output is drawn from a single transformer, the power rating of a PC PSU can range from 250 W to 1000 W.

PSUs are designed to ensure optimal system performance by maintaining a power rating that is **40 percent greater** than the calculated system load.



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Complete the lesson knowledge check



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Lesson knowledge check 1

Match the voltage on the left with the elements powered by DC voltage on the right.

1	+/- 12 V	A	The microprocessor, memory, and circuitry of the motherboard and internal peripherals
2	+/- 5	B	Motorized equipment and electromechanical peripherals such as cooling fans and secondary storage drives such as hard disk drives (HDDs), optical drives, and floppy drives

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Lesson knowledge check 1

Match the voltage on the left with the elements powered by DC voltage on the right.

1	+/- 12 V
2	+/- 5

A	Motorized equipment and electromechanical peripherals such as cooling fans and secondary storage drives such as hard disk drives (HDDs), optical drives, and floppy drives
B	The microprocessor, memory, and circuitry of the motherboard and internal peripherals

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Location

The physical element of a PSU is located near the rear of a PC chassis.

A cable connects the PSU to the wall socket (power main).

Commonly, a cooling fan is visible next to the power cord connection point.

Some PSUs have a power on and off switch in addition to the button commonly found on the front of the chassis.

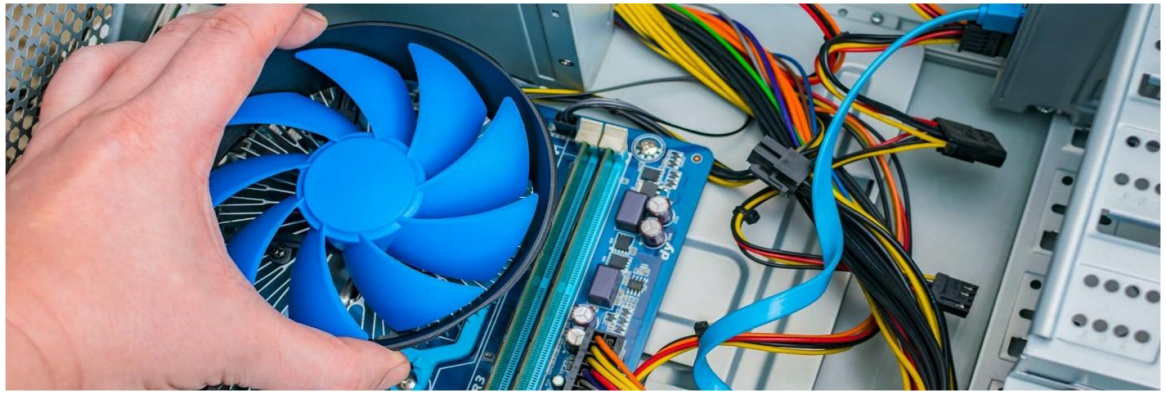


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Inside the system

Output cables

A PSU has output cables that provide power to the system motherboard and other internal peripheral devices.



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Inside the system (cont.)

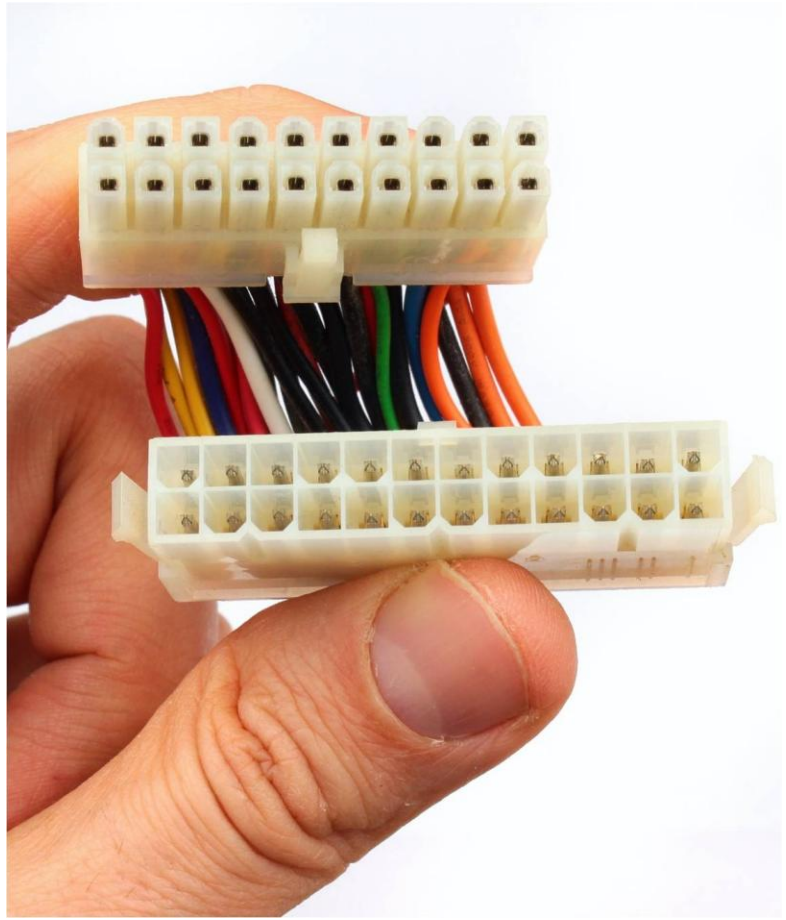
Connectors

The internal connectors usually have a:

- 20+4-pin connector that connects the PSU to the motherboard
- 4+4-pin connector used for powering graphical processing

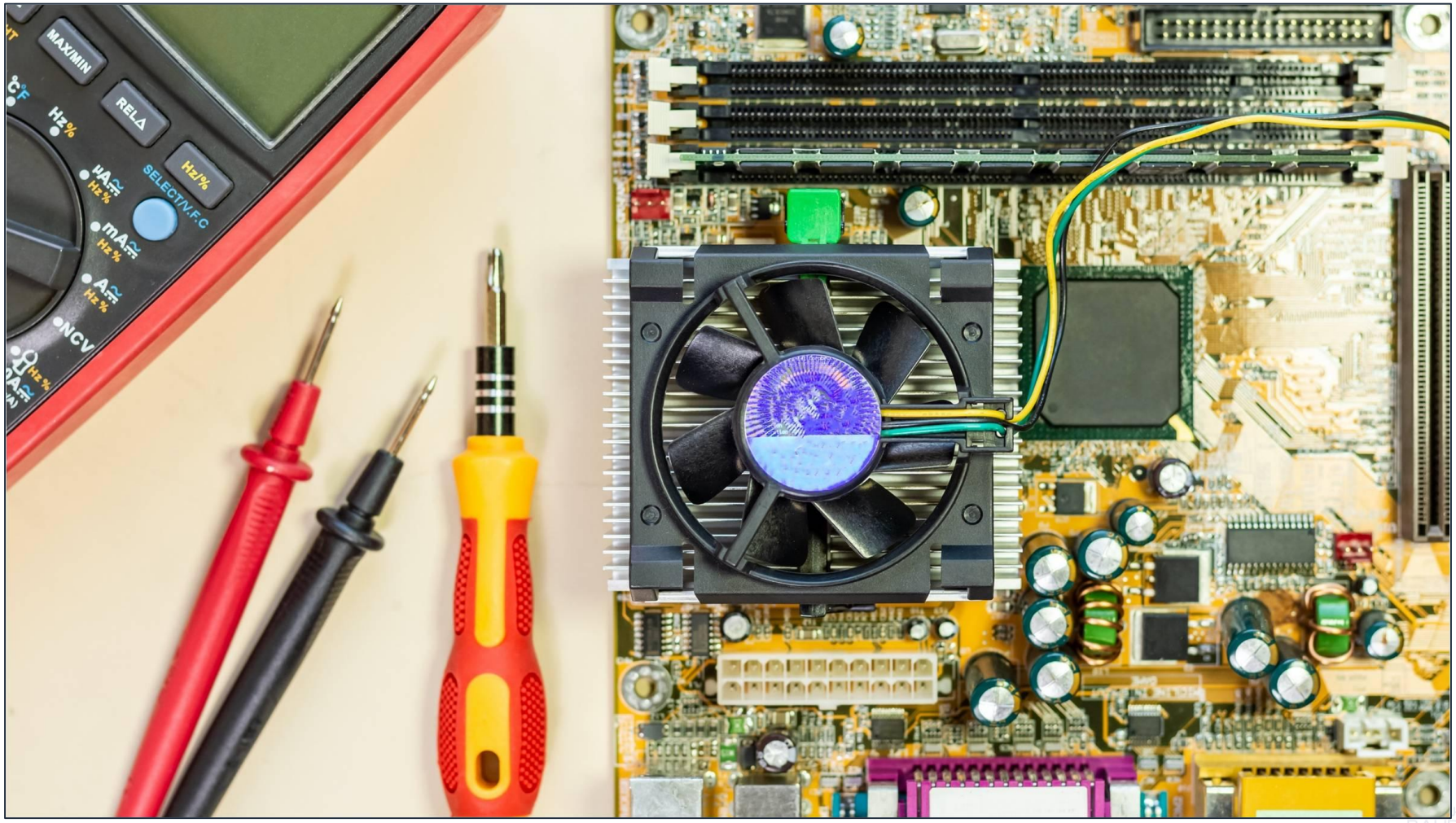
The PSU also has other assorted connectors for powering HDDs, optical drives, and other peripherals.

- These include cooling fans, if not provided on the motherboard.



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Lesson knowledge check 2

True or False: The motherboard uses the PSU 4+4-pin connector to draw its main source of power.

A. True

B. False

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Lesson knowledge check 2

True or False: The motherboard uses the PSU 4+4-pin connector to draw its main source of power.

A. True

B. False (Correct)

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Module 7: Power Supplies

Lesson 2: Install and remove a PSU

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Removing a PSU

A PSU is crucial to a PC and if faulty, needs to be replaced.



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Removing a PSU (cont.)

Steps to remove a PSU from a PC

1. Turn off the power.

Turn off the main power supply and disconnect the power cord from it.

Locate the PSU by following the main power cord to the point at which it connects to the PC tower.

Disconnect the power cord from the tower.



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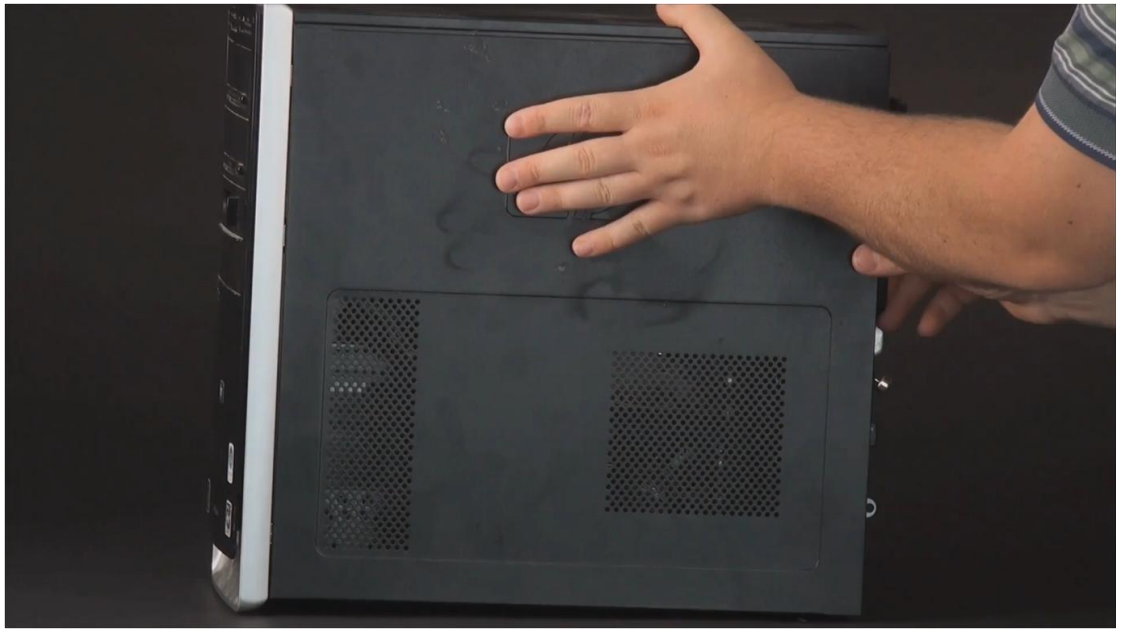
Removing a PSU (cont.)

Steps to remove a PSU from a PC

2. Remove the panel.

To remove the tower cover panel, remove the retaining screws, and slightly pull it back toward the rear of the computer tower.

Now, the PSU is visible.



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Removing a PSU (cont.)

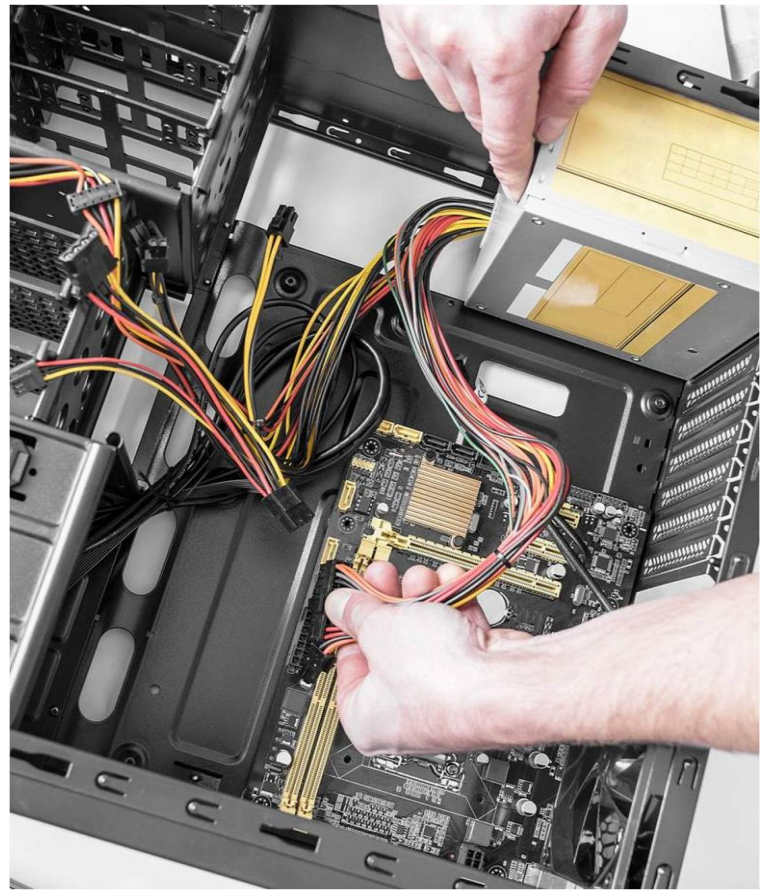
Steps to remove a PSU from a PC

3. Disconnect the PSU.

The output end of the PSU has several connectors leading to the system motherboard and other devices installed in the tower.

- Examples include the HDD and cooling fans.

Carefully unplug the PSU connections to each device.



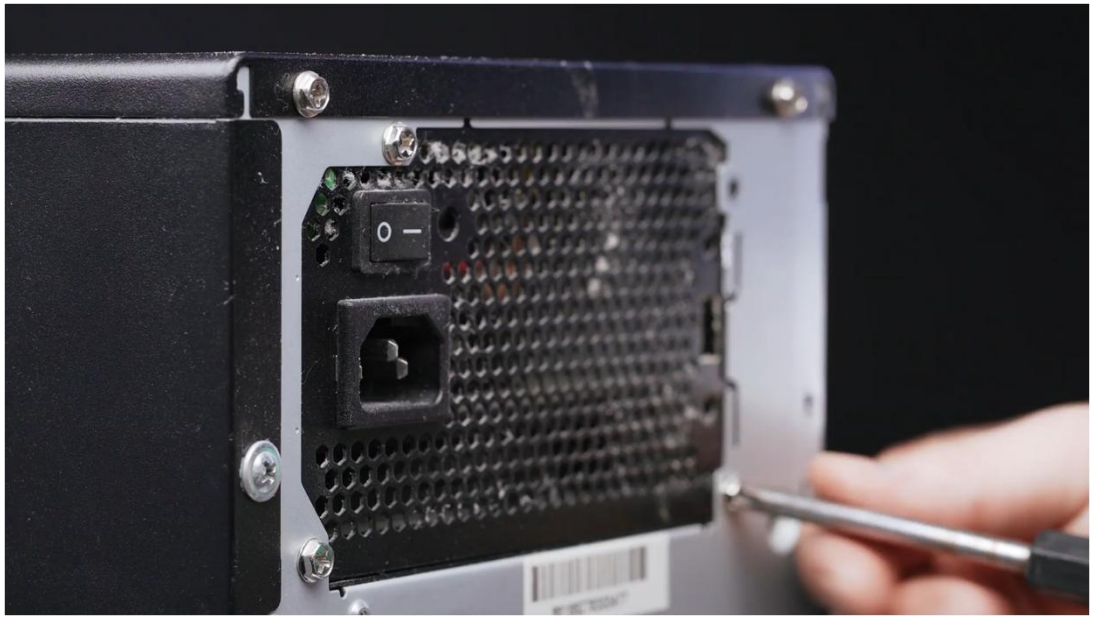
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Removing a PSU (cont.)

Steps to remove a PSU from a PC

4. Unscrew the PSU.

Remove the screws that mount the PSU to the tower.



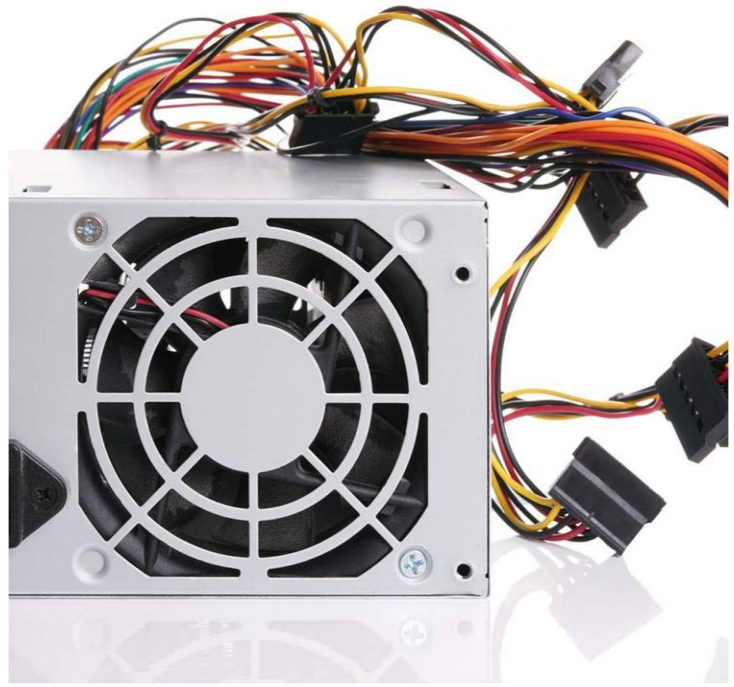
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Removing a PSU (cont.)

Steps to remove a PSU from a PC

5. Remove the PSU.

You can now safely remove the PSU from the PC.



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Complete the lesson knowledge check



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Lesson knowledge check 1

Match each power supply unit (PSU) removal step on the left with its description on the right.

1	Step 1
2	Step 2
3	Step 3
4	Step 4

A	Carefully unplug the connectors from the PSU to internal elements.
B	Turn off the main power, and disconnect all power cords and external connections.
C	Remove the PSU screws.
D	Open the tower panel by removing the screws.

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Lesson knowledge check 1

Match each power supply unit (PSU) removal step on the left with its description on the right.

1	Step 1
2	Step 2
3	Step 3
4	Step 4

B	Turn off the main power, and disconnect all power cords and external connections.
D	Open the tower panel by removing the screws.
A	Carefully unplug the connectors from the PSU to internal elements.
C	Remove the PSU screws.

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Installing a PSU

Steps to install a PSU in a PC



1. Insert the PSU.

Locate the PSU slot inside the tower, and insert the new PSU.



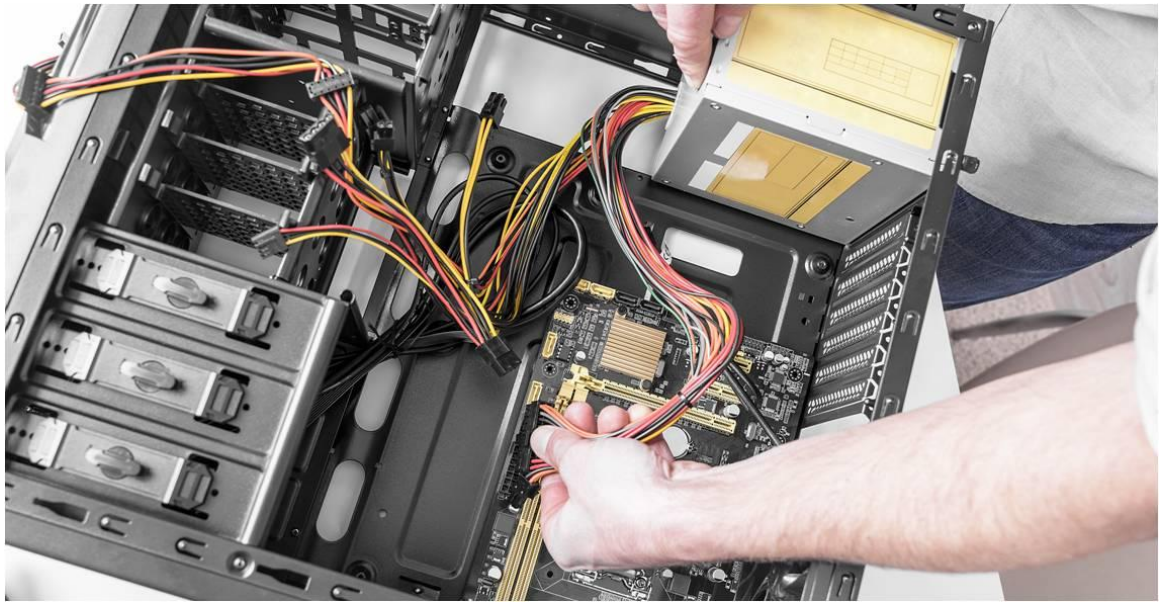
2. Secure the PSU.

Fasten the screws at the back of the tower tightly to secure the PSU.

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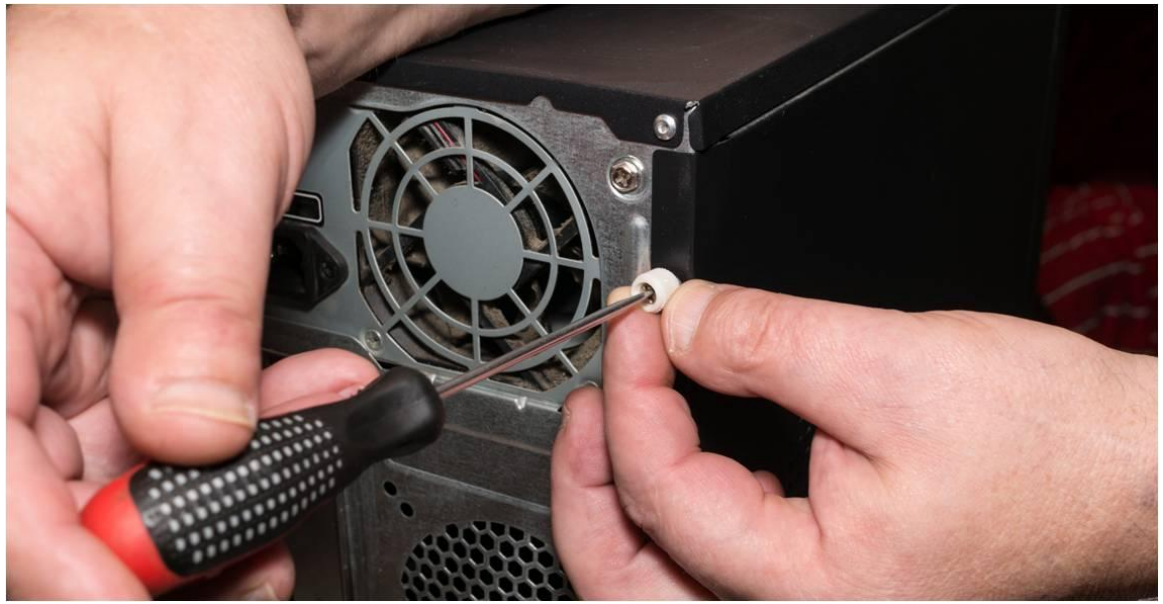
Installing a PSU (cont.)

Steps to install a PSU in a PC



3. Connect the PSU.

To connect the PSU, connect the PSU outputs to their respective internal devices, such as the motherboard, optical drive, and cooling fan.



4. Replace the panel.

Replace the panel on the side of the tower.

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Installing a PSU (cont.)

Steps to install a PSU in a PC



5. Connect the power cord.

Connect the power cord between the PSU and the main power source.



6. Reconnect the devices and cables.

Reconnect any external devices or cables.

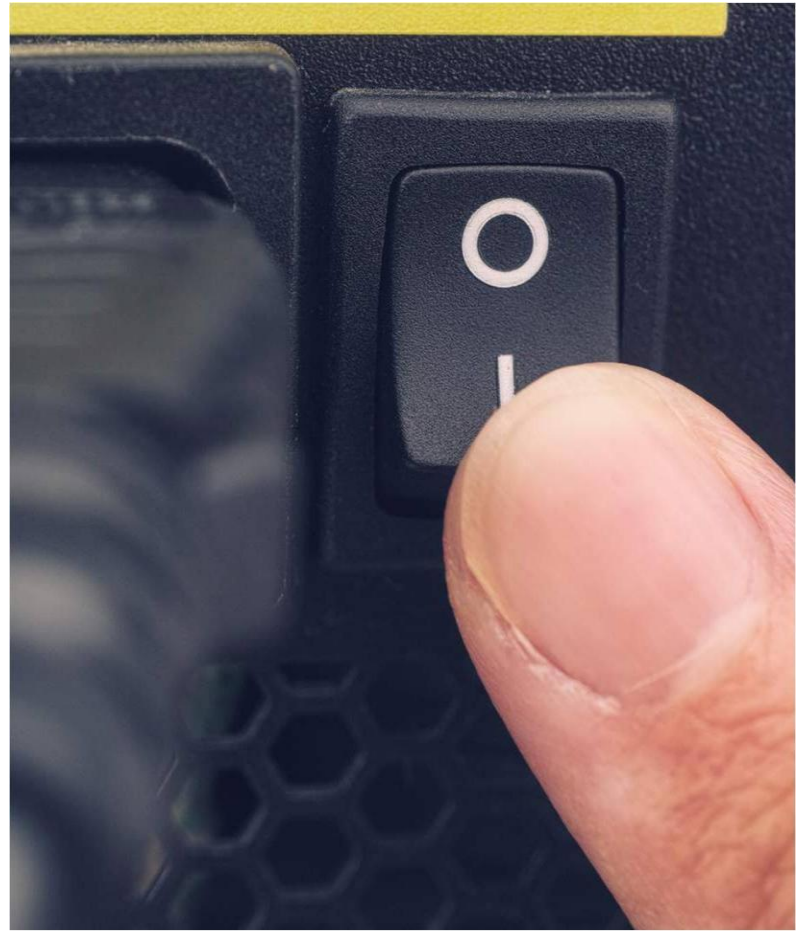
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Installing a PSU (cont.)

Steps to install a PSU in a PC

7. Test the PSU.

Test the installation of the PSU by turning on the main power source and pressing the power button on the tower to ensure that the PC turns on.



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Lesson knowledge check 2

Match each power supply unit (PSU) installation step on the left with its description on the right.

1	Step 1
2	Step 2
3	Step 3
4	Step 4
5	Step 5

A	Connect the PSU outputs to their respective internal devices, such as the motherboard.
B	Replace the panel on the side of the tower.
C	Locate the PSU slot inside the tower, and insert the new PSU.
D	Connect the power cord between the PSU and the main power source.
E	Fasten the screws at the back of the tower tightly to secure the PSU.

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Lesson knowledge check 2

Match each power supply unit (PSU) installation step on the left with its description on the right.

1	Step 1
2	Step 2
3	Step 3
4	Step 4
5	Step 5

C	Locate the PSU slot inside the tower, and insert the new PSU.
E	Fasten the screws at the back of the tower tightly to secure the PSU.
A	Connect the PSU outputs to their respective internal devices, such as the motherboard.
B	Replace the panel on the side of the tower.
D	Connect the power cord between the PSU and the main power source.

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Module 7: Power Supplies

Lesson 3: Troubleshoot a PSU

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Symptoms of PSU issues

Common PSU-related issues include:

- No power
- Unrequested restarts
- An unresponsive system
- Failure to detect internal devices (intermittently or permanently)
- Overheating
- Electrical charge on the computer case



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Lesson knowledge check 1

True or False: A symptom of a power supply unit (PSU) issue is a personal computer (PC) cooling fan that runs constantly.

A. True

B. False

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Lesson knowledge check 1

True or False: A symptom of a power supply unit (PSU) issue is a personal computer (PC) cooling fan that runs constantly.

A. True (Correct)

B. False

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How to troubleshoot suspected PSU issues

Troubleshooting checklist

Proceed through this troubleshooting checklist to investigate a suspected PSU-related issue.

1. Check the AC power connection and cord.

Check that the connection from the AC power input (from the main) to the PSU is firm. Make sure that the cord is not frayed or broken.

2. Check the DC power connection.

Check that the DC power connections inside the PC are firm. Examples include the motherboard, HDD, optical drive, and cooling fans.

3. Reseat and clean the connectors.

Reseat and clean the connectors. You might use a non-corrosive cleaner such as Isopropyl rubbing alcohol to clean the connectors.

4. Check the power voltage,

Check for the correct power voltage at various output points using a multimeter tool.

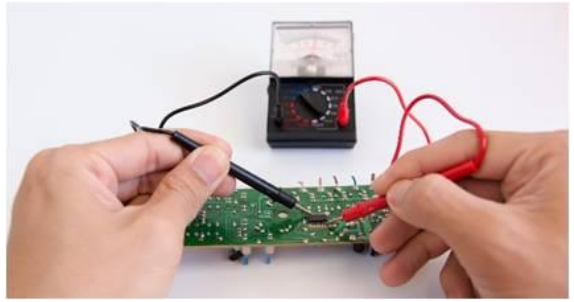


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How to troubleshoot suspected PSU issues (cont.)

Troubleshooting with the multimeter

Use the multimeter to check output points, or pins, to make sure that the correct voltage is being transmitted. There are multiple pins to check, each with its own required voltage.



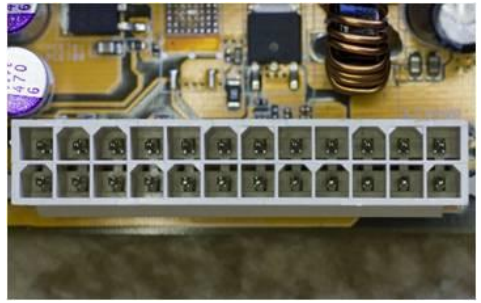
1. Attach the probe to PSU ground.

Attach the multimeter's black probe to the PSU ground port (also black).



2. Check the voltage.

Check the voltage of the orange *power-good* pin. The ideal voltage at power good is +3 V to +6 V.



3. Check the power at the other pins.

Check the power at each of the other pins. The pins on the male connector of the PSU and female connector on the motherboard might not be color coded.



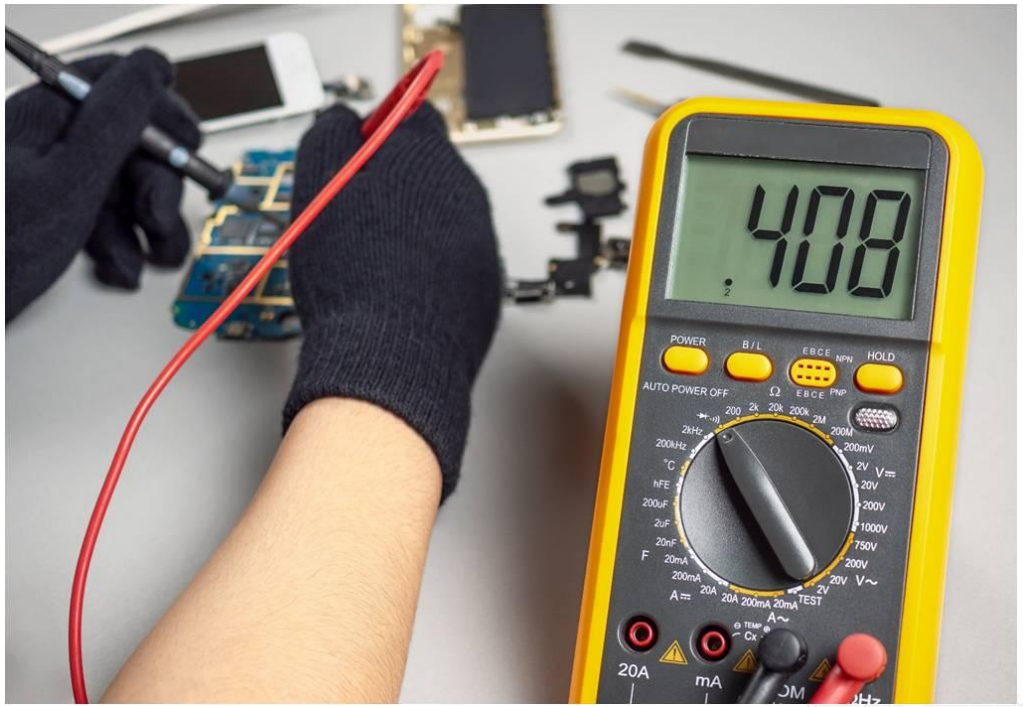
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How to troubleshoot suspected PSU issues (cont.)

Troubleshooting with the multimeter

This table outlines the voltage required for each pin. Variations of up to 5 percent are acceptable.

Pin	Voltage
Red	+5
White	-5
Yellow	+12
Blue	-12
Black	Ground (0)
Orange (also called power good	+3 to +6



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How to troubleshoot suspected PSU issues (cont.)

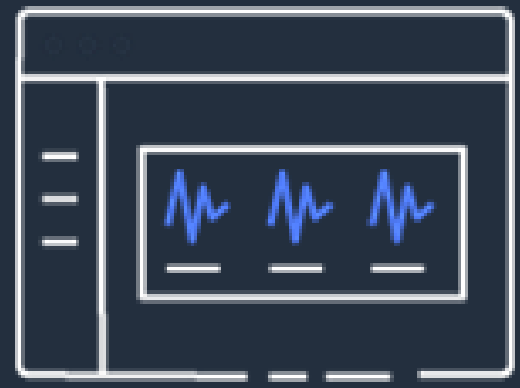
Power-good pin

The power-good pin is:

- P8-1 on an IBM PC/AT, Baby AT, and Low Profile eXtension (LPX) PSU motherboard connector
- Pin 8 on the Advanced Technology eXtended (ATX) PSU motherboard connector

To test the power-good voltage:

1. Connect the black probe of the multimeter to the black wire of the PSU connector (ground connection).
2. Connect the red probe of the multimeter to the orange wire of the PSU connector (power good)
3. Turn on the power supply, and check the reading on the multimeter.
   The reading should momentarily spike to +3 V or +6 V.



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How to troubleshoot suspected PSU issues (cont.)

Troubleshooting the PSU cooling fan

Cooling fans are critical to PSU operation because they help dissipate heat generated inside the PSU.

Cooling fan issues might be diagnosed when a system startup is followed by eventual malfunctions or burnout as the PSU circuit and components become warmer.

Cooling fans should always be checked for unexpected sounds and obstructions.

Modern ATX PSUs use brushless DC (BLDC) fans. This has reduced the incidence of mechanical failures. However, it does not reduce electrical issues.



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Complete the lesson knowledge check



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Lesson knowledge check 2

Which strategies are recommended for troubleshooting power supply unit (PSU)-related issues? (Select THREE.)

A. Check the alternating current (AC) power input connection to the PSU.
B. Check the direct current (DC) power connections inside the computer.
C. Turn on and off the computer.
D. Reseat and clean the connections.
E. Exchange the alternating current (AC) and direct current (DC) power connectors.
F. Remove the memory modules.

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Lesson knowledge check 2

Which strategies are recommended for troubleshooting power supply unit (PSU)-related issues? (Select THREE.)

A. Check the alternating current (AC) power input connection to the PSU. (Correct)

B. Check the direct current (DC) power connections inside the computer. (Correct)

C. Turn on and off the computer.

D. Reseat and clean the connections (Correct).

E. Exchange the alternating current (AC) and direct current (DC) power connectors.

F. Remove the memory modules.

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Module summary

In summary, in this module, you learned how to:

- Identify the purpose of a PSU
- Identify the physical elements of a PSU
- Identify the steps to remove a PSU from a PC
- Identify the steps to install a PSU in a PC
- Identify the symptoms of PSU-related issues
- Identify troubleshooting strategies for PSU-related issues

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Questions

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